Module 2 Data Modeling using ER Diagram

Learning objectives

- Know the key terminologies of ER Modeling;
- Draft an ER diagram using an ER Modelling Tool.

To start designing a database in an organization, we must understand their business and how they operate. As a database designer, you need to interview key positions such as manager, supervisor, or even the president of your client's organization to know what particular information they wish to maintain. Who and what particular role each member plays in the business process.

We must have a drawing board first so we can get a complete picture of the players in our database. That drawing board or tool is a simple pen and paper to draw your initial diagram: the entity-relationship diagram.

First, we need to define what an entity is.

An **entity** is an object, a thing, or an event that we want to keep track of in the organization.

In a school setup, STUDENT is an entity. FACULTY MEMBER is an entity. SCHEDULE is also an entity.

If you look in your Certificate of Matriculation (COM), there are quite a number of entities.

A particular entity has an identifiable characteristic. A STUDENT has a student number, last name, first name, gender, address. We call that characteristic an **attribute**.

If we draw that STUDENT entity with an attribute in an ERD, we use a rectangle to represent the entity with attributes on it.

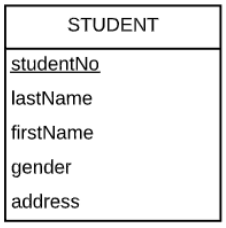


Figure 2-0 Entity Notation.

An attribute that is underlined represents a primary key. Primary key is used to uniquely identify an instance of an entity. Just like in real world applications, your student number is only on your own. No students share the same student number. Primary key is a unique identifier of an entity.

The school does not deal with STUDENTs alone. It maintains its own roster of teaching personnel known as FACULTY. And these two entities interact with each other.

FACULTY teaches STUDENTs.

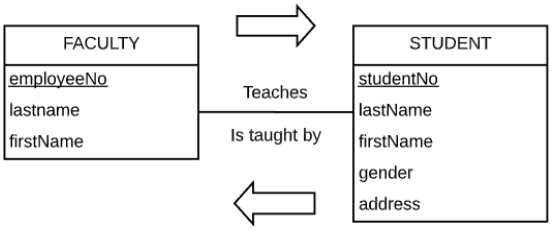


Figure 2-1 Entity Relationship Notation.

There lies the relationship between the two. We now have a complete set of entity-relationship diagrams. Relationship is indicated as a verb (or linking verb) that connects the two.

Reading the ER diagram can take both directions. Instead of just reading it from left to right, it can take the opposite. STUDENTs are taught by FACULTY.

Do we need to indicate relationships in both directions? I think not. One relationship will do. If the preferred reading is from left to right, there is no need to indicate the relationship from right to left.

Just take note of the notations used. Entity should be singular and in all caps. Relationships start with big letters. Primary key attribute is underlined.

Cardinality of Relationship

We can describe the relationship between two entities by means of its cardinalities. Cardinality shows the possible occurrences that may have between two entities.

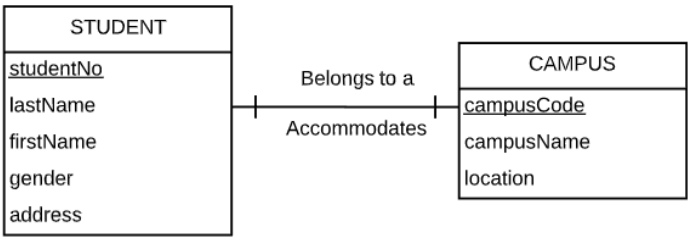


Figure 2-2 One-to-one relationship.

When the rule of the school is one STUDENT must only belong to one CAMPUS we show the notation one-to-one relationship with a bar symbol. It means that the relationship between the two restricts occurrences of only one student in a campus.

FACULTY on the other hand teaches not only one student but many students. We use the crow foot notation to signify many.

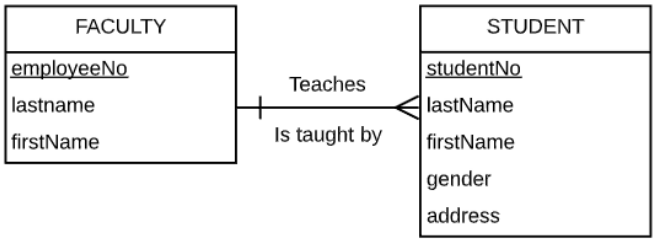


Figure 2-3 One-to-many or Many-to-one relationship.

But in real applications, we do not use just one faculty but many faculties that teach many students. So, we may change the notation from many-to-many.

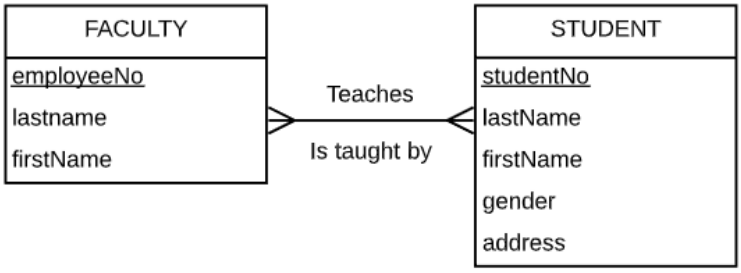


Figure 2-4 Many-to-many relationship.

Minimum and maximum cardinality

Cardinality is a rule. But sometimes your rule must be flexible enough so you can accommodate other possibilities. Our rule may adjust depending on a certain situation. We say, one faculty member teaches one student; one faculty teaches many students. Many faculty members teach one student; many faculty members teach many students.

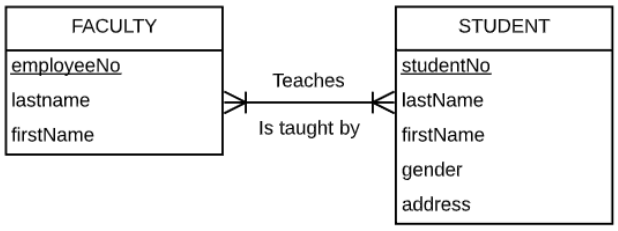


Figure 2-5 Minimum and maximum cardinality.

The minimum should be located in the outside of the cardinality notation; the maximum is located in the inside part.

We may also say that a FACULTY is not assigned to teach students. The Dean position is also a faculty member but he does not teach on certain occasions. To accommodate the dean position in our database, we must revise the notation about faculty and student relationship.

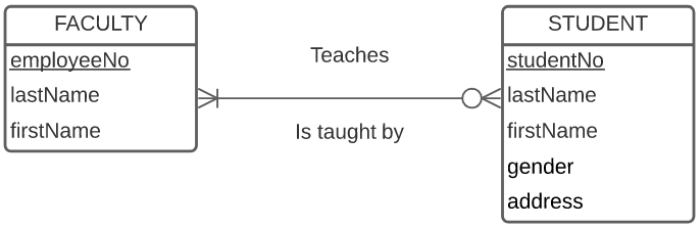
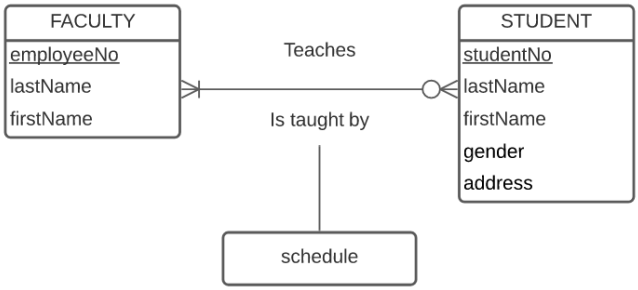


Figure 2-6 Zero or optional to many cardinalities.

We use the zero or the optional notation when we say that a faculty may not teach students. In our case, the dean does not teach students.

Gerund: An associative entity

We have the dilemma if we stick to the many-to-many relationship. If you look deeper in the relationship, where do the two entities meet? You are correct. You can clearly identify them in a certain schedule. But where will you put the schedule, in student or faculty? You can not put the schedule in the student entity because a student has many schedules. This will violate the rule of primary key. One occurrence of a student is allowed inside the entity. Same with the faculty. Even though faculty has many schedules, in doing so, it will violate the rule of primary key by putting too many employee numbers inside the entity.



Where do we put a schedule?

Figure 2-7 Confusion on where to put schedule attributes.

Because schedules can be placed neither of the two, we need to create a separate entity for the schedule. The entity is called the **gerund** or an associative entity. It serves as a bridge between two entities. Notice that SCHEDULE has other attributes present with other entities? These attributes connect other entities in a much clearer manner compared with the vague many-to-many relationships. Yes, you are correct! We get rid of the many-to-many relationships and convert it either to one-to-many, one-to-zero. Focus on these notations will be explained when we talk about relations.

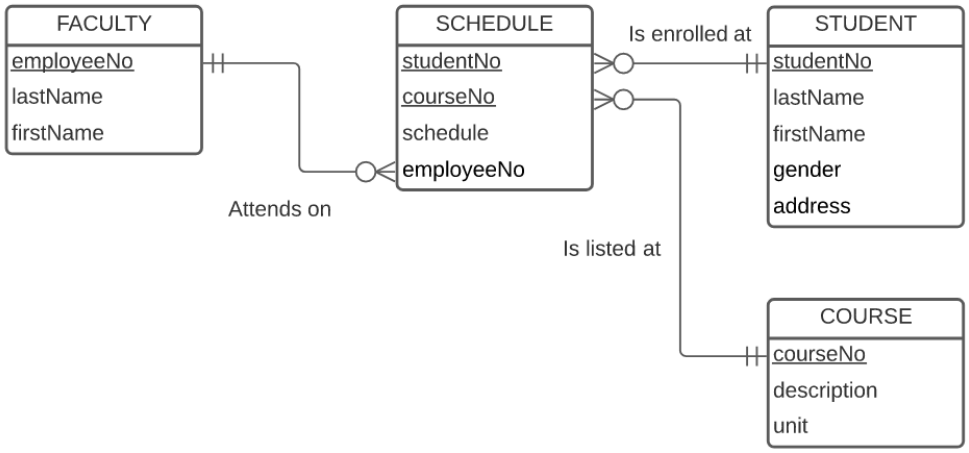


Figure 2-8 Separating schedule as an entity.

In some practice, if the minimum and maximum cardinality is one, only one bar is enough. With this notation, database designers know already that the minimum and maximum are the same. If we revise our diagram, we have this:

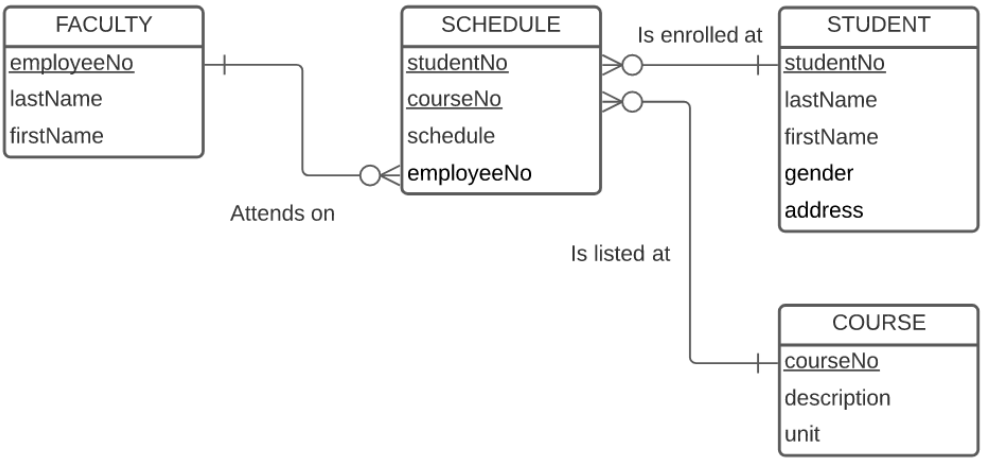


Figure 2-9 Minimum and maximum simplified into one bar.

The key takeaway here is, if you draw your initial ER diagram that would not be final. It is still subject for revisions and clarifications just like what we did in STUDENT and FACULTY entities. We will have more of these in discussions to come.

References

Gillension, M. (2012). Fundamentals of Database Management Systems. Retrieved from https://bookshelf.vitalsource.com

Hoffer, J. (2010). Modern Database Management (13th ed.). New York, NY: Pearson Education.

Rubias, J. (2016). Oo mahirap mag-aral, pero mas mahirap mag-paaral. Retrieved from https://joannerubias28.wordpress.com/2016/12/19/oo-mahirap-mag-aral-pero-mas-mahirap-mag-paaral/

Lucidchart. How Lucidchart works for you. Retrieved from https://www.lucidchart.com (2020)